

The One-Factor SD2P Projective-Tensor Question Was Answered Negatively

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Source question

Becerra Guerrero, López-Pérez, and Rueda Zoca, arXiv:1407.6038, prove in Corollary 3.6 that if Banach spaces X and Y both have the strong diameter two property (SD2P), then $X \widehat{\otimes}_\pi Y$ has SD2P. Immediately afterward they ask whether one SD2P factor is enough, and suggest $X \widehat{\otimes}_\pi \ell_2^2$ as a possible negative candidate. The question occurs on source PDF page 7.

The scanner’s earlier introductory “natural open question” is not a fresh target: it asks about stability from both factors and is answered by the same paper’s Corollary 3.6.

Later full negative answer

Langemets, Lima, and Rueda Zoca, arXiv:1609.02062, Corollary 3.9 (supporting PDF page 10), prove that for every $2 < p < \infty$ and $n \geq 3$,

$$\ell_\infty \widehat{\otimes}_\pi \ell_p^n \quad \text{and} \quad L_\infty[0, 1] \widehat{\otimes}_\pi \ell_p^n$$

do not have SD2P. On the other hand, $L_\infty[0, 1]$ has the Daugavet property and therefore has SD2P. Thus the second tensor product is an explicit counterexample to preservation from one factor.

Remark 3.11 of the answering paper (supporting PDF page 11) records the conclusion verbatim in mathematical substance: projective tensor products preserve SD2P when both factors have it, but not in general when only one does. This fully and negatively resolves the source question.

References

- J. Becerra Guerrero, G. López-Pérez, and A. Rueda Zoca, *Octahedral norms in spaces of operators*, arXiv:1407.6038.
- J. Langemets, V. Lima, and A. Rueda Zoca, *Octahedral norms in tensor products of Banach spaces*, arXiv:1609.02062.